

Part 1 (Linear Algebra)

Exercise 1Answer to the Q: No 1

Writing the system as $Ax = b$,

$$A = \begin{pmatrix} 2 & 3 & 1 \\ 4 & 7 & 5 \\ 0 & -2 & 2 \end{pmatrix}, \quad x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}, \quad b = \begin{pmatrix} 8 \\ 20 \\ 0 \end{pmatrix}$$

It can be expressed as,

$$\begin{pmatrix} 2 & 3 & 1 \\ 4 & 7 & 5 \\ 0 & -2 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 8 \\ 20 \\ 0 \end{pmatrix}$$

Answer to Q: No 2

Solving for x ,
Augmented matrix equivalent to this system,

$$\left[\begin{array}{ccc|c} 2 & 3 & 1 & 8 \\ 4 & 7 & 5 & 20 \\ 0 & -2 & 2 & 0 \end{array} \right]$$

Using Gaussian elimination method,
we can solve this ~~eqn~~ matrix.

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$$R_2 = (r_2 - 2r_1)$$

$$\left[\begin{array}{ccc|c} 2 & 3 & 1 & 8 \\ 0 & 1 & 3 & 4 \\ 0 & -2 & 2 & 0 \end{array} \right]$$

$$R_3 = (r_3 + 2r_2)$$

$$\left[\begin{array}{ccc|c} 2 & 3 & 1 & 8 \\ 0 & 1 & 3 & 4 \\ 0 & 0 & 8 & 8 \end{array} \right]$$

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From the matrix, we can find

$$8x_3 = 8$$

$$\therefore x_3 = 1$$

$$x_2 + 3x_3 = 1$$

$$\Rightarrow x_2 + 3 \times 1 = 1$$

$$\therefore \Rightarrow x_2 = -2$$

$$2x_1 + 3x_2 + x_3 = 8$$

$$\Rightarrow 2x_1 + 3 \times (-2) + 1 = 8$$

$$\Rightarrow 2x_1 - 6 + 1 = 8$$

$$\therefore x_1 = 2$$

$$\therefore x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$$

this is the only unique solution possible in this case.



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~~Part 2~~

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~~Answer to Q: No 1~~

Exercise 2

Answer to Q: No 1

Given, $A = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}$

we know,

$$\det(A - \lambda I) = 0$$

Here, $A - \lambda I = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

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$$\therefore \det \begin{bmatrix} 1-\lambda & -1 & 0 \\ -1 & 2-\lambda & -1 \\ 0 & -1 & 1-\lambda \end{bmatrix} = 0$$

Here, $(1-\lambda)[(2-\lambda)(1-\lambda) - (-1)(-1)]$
 $- 1[(-1)(1-\lambda) - 0] + 0 = 0$

$$\Rightarrow (1-\lambda)(\lambda^2 - 3\lambda + 2) + 1(\lambda - 1) = 0$$

$$\Rightarrow (1-\lambda)(\lambda^2 - 3\lambda + 2) - (1-\lambda) = 0$$

$$\Rightarrow (1-\lambda) [(\lambda - 3\lambda + 1) - 1] = 0$$

$$\Rightarrow (1-\lambda) (\lambda - 3\lambda) = 0$$

$$\Rightarrow \lambda (1-\lambda) (\lambda - 3) = 0$$

$$\therefore \lambda = 0$$

$$1 - \lambda = 0$$

$$\Rightarrow \lambda = 1$$

$$\therefore \lambda - 3 = 0$$

$$\Rightarrow \lambda = 3$$

$\therefore$ The eigenvalues of A are

0, 1, 3

[shown]

Answer to Q. 2

From previous solution, we
get eigenvalues are 0, 1, 3

Let $x = [x_1, x_2, x_3]$ be the
eigen vectors corresponding to
eigen value λ

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we will consider,

$$[A - \lambda I]x = 0$$

to solve eigenvectors.

$$\therefore \begin{bmatrix} 1-\lambda & -1 & 0 \\ -1 & 2-\lambda & -1 \\ 0 & -1 & 1-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad (1)$$

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when, $\lambda = 0$,

equation (i) $\Rightarrow$

$$\begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow x_1 - x_2 = 0 \therefore x_1 = x_2$$

$$\Rightarrow -x_1 + 2x_2 - x_3 = 0 \therefore x_3 = 2x_2 - x_1$$

$$\therefore x_3 = x_2$$

$\therefore$ we get, $x_1 = x_2 = x_3$

Let, $x_1 = 1 \therefore x_1 = x_2 = x_3 = 1$

$$\therefore \text{for } \lambda = 0, x = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

when, $\lambda = 1$, equation (i) become,

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 1 & -1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$2) -x_2 = 0 \quad \therefore x_2 = 0$$

$$\Rightarrow -x_1 + x_2 - x_3 = 0 \Rightarrow -x_1 = x_3$$

$$\text{Let, } x_1 = 1, x_3 = -1$$

$$\therefore \text{for } \lambda = 1, x = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

when, $\lambda = 3$, equation (i) become,

$$\begin{bmatrix} -2 & -1 & 0 \\ -1 & -1 & -1 \\ 0 & -1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

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$$\Rightarrow -2x_1 - x_2 = 0 \quad \therefore -2x_1 = x_2$$

$$\Rightarrow -x_1 - x_2 - x_3 = 0$$

$$\Rightarrow -x_1 + 2x_1 - x_3 = 0$$

$$\Rightarrow x_1 = x_3$$

$$\text{Let, } x_1 = 1, x_2 = -2, x_3 = 1$$

$$\therefore \text{ for } \lambda = 3, x = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

Answer to Q: No 3

A is diagonalizable, as it has three distinct eigenvalues.

$\therefore$ The matrix A is diagonalized as

$$A = PDP^{-1}; D = \text{Diagonal matrix contains eigenvalues}$$

where, $P = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & -2 \\ 1 & -1 & 1 \end{pmatrix}$ ^{Relevant}, $D = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}$

P contains corresponding eigenvectors.

Part 2 (Probability and Statistics)

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Exercise 3

given, $P(X=1)=0.2$, $P(X=2)=0.3$,

$$P(X=3)=0.4, P(X=4)=0.1$$

we know, the mean of a discrete random variable,

$$E(X) = \sum x P(X=x)$$

$$= 1 \times 0.2 + 2 \times 0.3 +$$

$$3 \times 0.4 + 4 \times 0.1$$

$$= 2.4 \text{ (Answer)}$$

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the variance,

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$\text{Here, } E(X^2) = 1^2(0.2) + 2^2(0.3) + 3^2(0.4) + 4^2(0.1)$$

$$= 0.2 + 1.2 + 3.6 + 1.6$$

$$= 6.6$$

$$\therefore \text{Var}(X) = 6.6 - (2.4)^2$$

$$= 6.6 - 5.76$$

$$= 0.84$$

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(Answer)

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Exercise 4

① By using linearity of expectation property, we get,

$$E[aX+b] = aE[X] + b$$

$$\therefore E[2X_1+3] = 2E[X_1] + 3$$

$$= 2 \times 2 + 3$$

$$\therefore E[2X_1+3] = 7$$

② By using the property of variance,

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we get, ~~$V[aX+b] = aV[X]$~~ $V[aX+b] = a^2 V[X]$

$$\therefore V[2X_1+3] = 2^2 V[X_1]$$

[adding a constant doesn't change its spread]

$$= 2^2 \times 1$$

$$= 4 \quad (\text{Answer})$$

③ As X_1, X_2, X_3 are independent random variables, their covariance is zero. The correlation of any variable with itself is always 1.

$\therefore$ The correlation matrix is the 3×3 identity matrix:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{aligned} \textcircled{4} \quad E[X_1 + X_2 + X_3] \\ &= E[X_1] + E[X_2] + E[X_3] \\ &= 2 + 3 + 0 \\ &= 5 \end{aligned}$$

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$$(5) V[X_1 + X_2 + X_3]$$

$$= V[X_1] + V[X_2] + V[X_3] \quad [\because \text{variables are independent, the sum of the variances is the sum of individual variances}]$$

$$= 1 + 2 + 3 = 6 \quad (\text{Ans})$$

$$(6) E\left[\frac{X_1 + X_2 + X_3}{3}\right]$$

$$= \frac{1}{3} E[X_1 + X_2 + X_3]$$

$$= \frac{1}{3} \times 5 \quad [\because E[X_1 + X_2 + X_3] = 5 \text{ from question 4}]$$

$$= \frac{5}{3}$$

(Answer)

$$\textcircled{4.} \quad \sqrt{\frac{X_1 + X_2 + X_3}{3}}$$

$$= \left(\frac{1}{3}\right)^{\frac{1}{2}} \sqrt{X_1 + X_2 + X_3}$$

[Using the property
of variance]

$$= \frac{1}{9} \sqrt{X_1 + X_2 + X_3}$$

$$= \frac{1}{9} \times 6 \quad [\text{got from question 5}]$$

$$= \frac{6}{9}$$

$$= \frac{2}{3} \quad (\text{Answer})$$

Exercise 5Ans to Q: No 1

Given, $Y = X^{\sim}$

if $y \geq 0$, for $Y \leq y$.

$$X^{\sim} \leq y \Rightarrow -\sqrt{y} \leq X \leq \sqrt{y}$$

Answer to Q: No 2

Given $F_Y(y) = P(Y \leq y)$ [Definition of cdf]

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$$\Rightarrow F_Y(y) = P(X^{\sim} \leq y) [\because Y = X^{\sim}]$$

$$\therefore F_Y(y) = P(-\sqrt{y} \leq X \leq \sqrt{y})$$

Answer to Q: No 3

[Theorem]

As F_X is the cdf of X , we can apply the probability of a cdf on continuous random variable such as.

$$P(a \leq X \leq b) = F_X(b) - F_X(a)$$

using this ~~idea~~ ^{to} the interval
we got from question 2,

$$F_Y(u) = P(-\sqrt{u} \leq X \leq \sqrt{u})$$

$$\therefore F_Y(u) = F_X(\sqrt{u}) - F_X(-\sqrt{u})$$

Answer to the Q: No 4

We got from the property of
cdf of normal distribution,

$$F_X(-x) = 1 - F_X(x)$$

~~Putting this~~ Applying this to the
equation we got from question 3, ²

$$F_Y(u) = F_X(\sqrt{u}) - [1 - F_X(\sqrt{u})]$$

$$= F_X(\sqrt{u}) - 1 + F_X(\sqrt{u})$$

$$\therefore F_Y(u) = 2F_X(\sqrt{u}) - 1$$



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[deduced]

Answer to the Q: No 5

we got from question 4,

$$F_Y(y) = 2 F_X(\sqrt{y}) - 1$$

For $P(Y < 3)$:

$$\begin{aligned} P(Y < 3) &= F_Y(3) \\ &= 2 F_X(\sqrt{3}) - 1 \\ &= 2 F_X(1.732) - 1 \\ &= 2 \times 0.9582 - 1 \end{aligned}$$

[$F_X(1.732)$
in a standard
normal table is
approx 0.9582]

$$\begin{aligned} &= 1.9164 - 1 \\ &= 0.9164 \end{aligned}$$

for $P(Y < 5)$:

$$P(Y < 5) = F_Y(5)$$

$$= 2F_X(\sqrt{5}) - 1$$

$$= \cancel{2 \times 2.236} - 1$$

$$= 2F_X(2.236) - 1$$

$$= 2 \times 0.9873 - 1$$

[in the standard
normal table

$$F_X(2.236) \approx 0.9873$$

$$= 1.9746 - 1$$

$$= 0.9746$$

(Answer)

Answer to Q: No. 6

The pdf of X , $f_X(y)$ is
the first derivative of cdf,
 $F_X(y)$; which is

$$f_X(y) = \frac{d}{dy} F_X(y) \\ = F'_X(y)$$

Answer to Q: No. 7

From, Question 4, we got,

$$F_X(y) = 2 F_X(\sqrt{y}) - 1$$

Differentiating this equation, we get

$$f_X(y) = \frac{d}{dy} [2 F_X(\sqrt{y}) - 1]$$

$$\Rightarrow f_X(y) = 2 \cdot \frac{d}{dy} [F_X(\sqrt{y})]$$

$$\begin{aligned}
 &= 2 \cdot f'_x(\sqrt{y}) \cdot \frac{d}{dy}(\sqrt{y}) \\
 &= 2 \cdot f'_x(\sqrt{y}) \cdot \frac{1}{2\sqrt{y}} \left[\because f'_x(x) \right. \\
 &= \left. \frac{1}{\sqrt{y}} f'_x(\sqrt{y}) = f'_x(x) \right]
 \end{aligned}$$

$$\therefore f_y(y) = \frac{1}{\sqrt{y}} f'_x(\sqrt{y})$$

[Deduced]

Answer to Q: No 8

given, $f_x(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right)$

putting $x = \sqrt{y}$,

$$\begin{aligned}
 \Rightarrow f_x(\sqrt{y}) &= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(\sqrt{y})^2}{2}\right) \\
 &= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{y}{2}\right)
 \end{aligned}$$

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for

putting $f_X(\sqrt{y}) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{y}{2}\right)$

in equation we get in question 7,

$$f_Y(y) = \frac{1}{\sqrt{y}} \left[\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{y}{2}\right) \right]$$

$$\therefore f_Y(y) = \frac{1}{\sqrt{2\pi y}} \exp\left(-\frac{y}{2}\right)$$

[Deduced]

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Exercise 6Answer to Q: No 1

Here, $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$

$$\therefore E[\bar{X}_n] = E\left[\frac{1}{n} \sum_{i=1}^n X_i\right]$$

$$= \frac{1}{n} E\left[\sum_{i=1}^n X_i\right]$$

* [The expected value (mean) is a linear operator, we can opt out the constant]

$$= \frac{1}{n} \sum_{i=1}^n E[X_i]$$

$$= \frac{1}{n} \sum_{i=1}^n \mu [E[X_i] = \mu]$$

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$$= \frac{1}{n} (\mu + \mu + \dots + \mu)$$

$$= \frac{1}{n} (n\mu)$$

$$\therefore E[\bar{X}_n] = \mu$$

Answer to Q: No 2

$$\text{given, } \bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

$$\therefore V[\bar{X}_n] = V\left[\frac{1}{n} \sum_{i=1}^n X_i\right]$$

$$= \frac{1}{n^2} V\left[\sum_{i=1}^n X_i\right]$$

$$[\because V(aZ) = a^2 V(Z)]$$

$$= \frac{1}{n^2} \sum_{i=1}^n V[X_i]$$

$$= \frac{1}{n^2} \sum_{i=1}^n \sigma^2 \quad [\because V[X_i] = \sigma^2]$$



$$= \frac{1}{n} \cdot n\sigma^2$$

$$\therefore V[\bar{X}_n] = \frac{\sigma^2}{n} \quad (\text{Answer})$$

Answer to Q: No 3

As each X_i is normally distributed and independent,

their sum is also ~~normal~~ normally distributed.

$$\therefore \sum_{i=1}^n X_i \sim N(n\mu, n\sigma^2)$$

$\therefore \bar{X}_n$ is a linear combination of X_i variables,

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

$$\therefore \bar{X}_n \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

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Module 3(Python Programming)

Exercise 1

```
human_years = float(input("Enter human years: "))

if human_years < 0:
    print("Error: Age cannot be negative.")

elif human_years <= 2:
    dog_years = human_years * 10.5
    print("Dog years =", dog_years)  #each of the first two human years =
10.5 dog years,
    # then each additional human year = 4 dog years
    #2 year → 21
    #5 year → 21 + (3×4) = 33

else:
    dog_years = 21 + (human_years - 2) * 4
    print("Dog years =", dog_years)
```

Sample output:

```
Enter human years: -5
Error: Age cannot be negative.
```

Exercise 2

```
import math

a = float(input("Enter value of a: "))
b = float(input("Enter value of b: "))
c = float(input("Enter value of c: "))

discriminant = b**2 - 4*a*c

if discriminant < 0:
    print("The equation has no real roots.")

elif discriminant == 0: #[Discriminant D=b2-4ac]
    root = -b / (2*a)
    print("The equation has one real root.")
    print("Root =", root)
```



```

else:
    root1 = (-b + math.sqrt(discriminant)) / (2*a)
    root2 = (-b - math.sqrt(discriminant)) / (2*a)

    print("The equation has two real roots.")
    print("Root 1 =", root1)
    print("Root 2 =", root2)

```

Sample output:

```

Enter value of a: 1
Enter value of b: -5
Enter value of c: 6
The equation has two real roots.
Root 1 = 3.0
Root 2 = 2.0

```

Exercise 3

```

def edit_distance(s, t):

    # If first string is empty
    if len(s) == 0:
        return len(t)

    # If second string is empty
    elif len(t) == 0:
        return len(s)

    else:
        cost = 0

        if s[-1] != t[-1]:
            cost = 1

        # Recursive calls
        d1 = edit_distance(s[:-1], t) + 1
        d2 = edit_distance(s, t[:-1]) + 1
        d3 = edit_distance(s[:-1], t[:-1]) + cost

        # Return minimum distance
        return min(d1, d2, d3)

# Main program

```

```

string1 = input("Enter first string: ")
string2 = input("Enter second string: ")

distance = edit_distance(string1, string2)

print("Edit distance =", distance)

```

Sample output:

```

Enter first string: kitten
Enter second string: sitting
Edit distance = 3

```

Exercise 4

#if the sum of the angels is 180, then its a triangle , otherwise false

```

class Triangle:

    # Constructor
    def __init__(self, angle1, angle2, angle3):
        self.angle1 = angle1
        self.angle2 = angle2
        self.angle3 = angle3

    # Number of sides
    self.number_of_sides = 3

    def check_angles(self):

        if (self.angle1 + self.angle2 + self.angle3) == 180:
            return True
        else:
            return False

my_triangle = Triangle(90, 30, 60)

print("Number of sides =", my_triangle.number_of_sides)
print("Angles form a triangle?", my_triangle.check_angles())

```

Sample output:

Number of sides = 3
Angles form a triangle? False